Controlling data selections Activity Book

Create Visualizations with Qlik Sense

CVQS May 2023

© 2023 QlikTech International AB. All rights reserved. All company and/or product names may be trade names, trademarks and/or registered trademarks of the respective owners with which they are associated.

Table of Contents

Controlling data selectionsActivity Book

Table of Contents

Activity 1: Inserting set analysis with expression editor

Challenge 5
Hints 5
Solution 6

Activity 2: Working with set analysis

Challenge 7
Hints 7
Solution 8

Activity 3: Working with alternate states

Challenge 9
Hints 9
Solution 12

(Optional) Challenge 4: Validating set expressions

Challenge 13
Hints 13
Solution 14

(Optional) Activity 5: Combining set analysis and bookmarks

Challenge 15
Hints 15
Solution 17

Activity 1: Inserting set analysis with expression editor

Summary:

- Use the expression editor panel to define set analysis based on selections.
- Avoid typing the expression.
- Use set expressions out of aggregations.

Qlik Sense app: ABC Sales Dashboard v1

Challenge

Create a new KPI object in the sheet *Sales Analysis*, in the **ABC Sales Dashboard v1** app.

- The object will display the *Americas* Avg deal size.
- The expression will not be affected by user selections.

Note: Measures and data shown depend on today's date. If you reload your app data, your results might differ from the solution image results.

Hints

- Copy and paste the *"Avg deal size*"*, as this is very similar to the object we want achieve. Then, modify the expression.
- Note that it is important to place the cursor before the expression once you are going to inset the set analysis.

Step-by-step guide

1. In the **ABC Sales Dashboard v1** app, go to the sheet called *Sales Analysis* and click **Edit sheet**.
2. Copy the *"Avg deal size*"* object and paste it below the *"Current Year H1"* button.
3. Come back to analysis mode and select the *"Americas"* button on the left to apply this selection.
4. When the selection is applied to the app, under **Data>Measures>Avg deal size**, click the button to **Unlink measure**.
5. **Open expression editor** and look at the right side of the window. Under **Set expressions**, select **Use current selections** and click **Insert**. Confirm that the set expression has been inserted ahead of the existing expression:

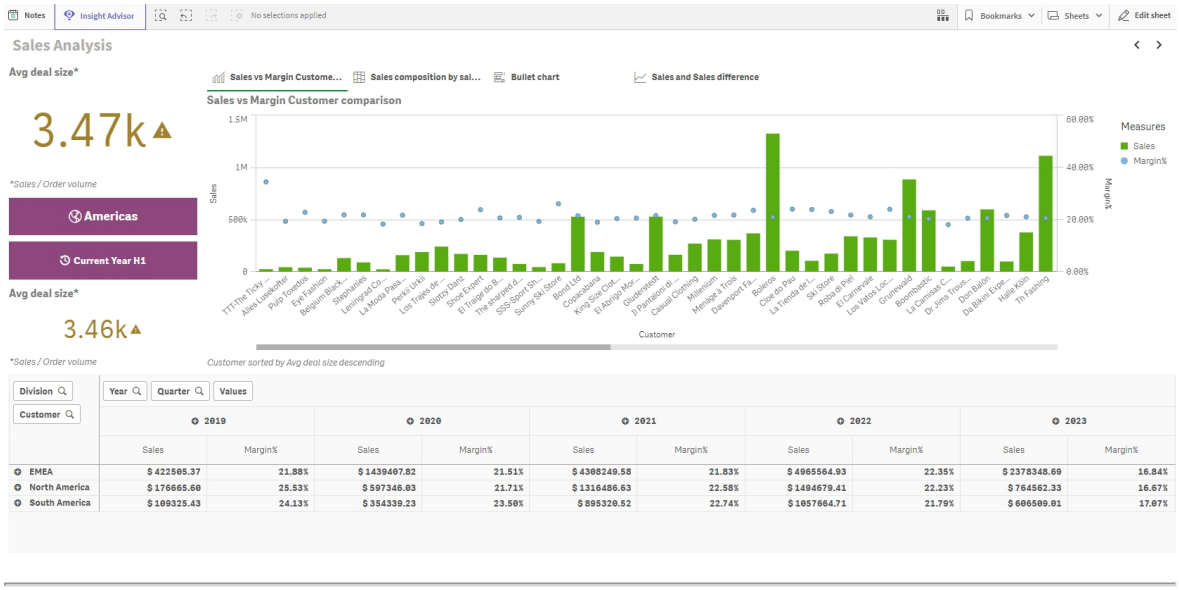
   ```
   {<Division={'North America','South America'}>}[Sales]/[Num Orders]
   ```

   We are using an **outer set analysis**, as the set expression is outside the aggregations, because master measures are applied in the expression. It is equivalent to this inner set analysis:

```
Sum({<Division={'North America','South America'}>}Sales)/Count({<Division={'North America','South America'}>}distinct OrderID)
```

6. Click **Apply** and **Edit sheet**. Note that the result of the KPI object that we have just created is the same as the other above, because we applied the Americas button selection to create the set expression.
7. If you clear selections, you will confirm that the Avg deal size for America is a little bit lower that the result displayed in the first KPI.

Solution



Activity 2: Working with set analysis

Summary:

- Explore and work with the Set analysis syntax in expressions.
- Use Set analysis in the visualizations in the app.

Qlik Sense app: ABC Sales Dashboard v1

Challenge

Create a new sheet, *Regional Orders*, in the **ABC Sales Dashboard v1** app and display the monthly order volume trend for each division.

- Add the following fields as filters in the sheet: *Year, Quarter, Month, Division*, and *Customer*
- Show the EMEA division monthly order volume using a line chart
- Show the Americas monthly order volume in another line chart with one line for each division in the Americas region
- Show value labels and data points in the line charts

Note: Measures and data shown depend on today's date. If you reload your app data, your results might differ from the solution image results.

Hints

- As the charts you create will be very similar, you can create one and copy it, then make changes to the measures and expressions.
- The set analysis statement is declared with curly brackets, inside the function.
  - The **$** set identifier represents the current selections.
  - The set modifier specifies the field values. For instance: *Division={'EMEA'}*.
- Outer set expression can be used instead of inner. In that case, the set analysis statement must be place ahead of the function.

Step-by-step guide

1. In the **ABC Sales Dashboard v1** app, create a new sheet called *Regional Orders* and open it in edit mode.
2. Add the following fields in a filter pane in the sheet: *Year, Quarter, Month, Division*, and *Customer*.

Create the EMEA Line chart

3. Create a line chart using *Month* as dimension and *Num Orders* as measure. Then make sure the *Month* dimension is set to **Sort numerically ascending** in the **Sorting** area and that the **Include null values** box is unchecked.
4. Use the properties panel to **Show data points** and display **Value labels**, under **Appearance > Presentation**.

5. Modify the *Num Orders* measure by breaking the link to the master item and using the following expression:

```
Count({$<Division={'EMEA'}>} distinct OrderID)
```

6. Add a title to the chart, *Order volume trend - EMEA*.

Create the Americas Line chart

7. Copy and paste the existing line chart. You will now have two line charts that are the same.
8. Select one of the line charts, and change the following:
   - Replace the measure using the following expression:

     ```
     Count({$<Division={"*America"}>} distinct OrderID)
     ```

   - Change the title, *Order volume trend - Americas*
   - Add a second dimension: *Division*
9. Under **Appearance > Colors and legend**, clear **Show legend title**. Then, position legend at the top.

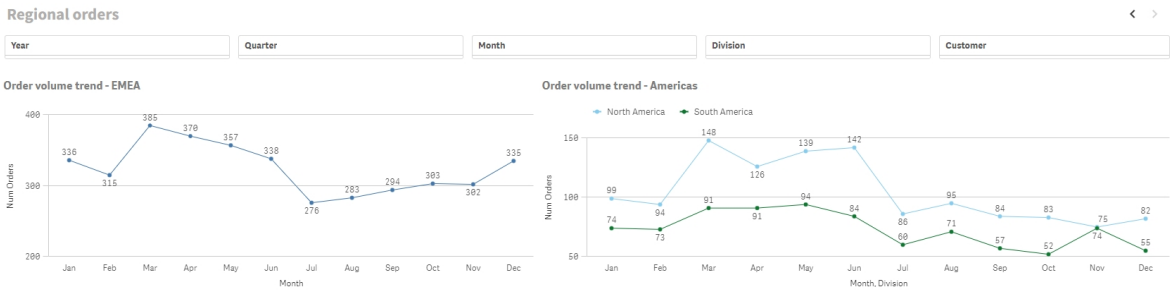
   Question: What if you also want to display the total trend line for Americas? In this case, you could transform the chart as follows:

   Remove the second dimension (*Division*) from the chart and create two more measures, one for North America and one for South America:

   - ```
     Count({$<Division={'North America'}>} distinct OrderID)
     ```

   - ```
     Count({$<Division={'South America'}>} distinct OrderID)
     ```

Solution



Activity 3: Working with alternate states

Summary:

- Explore Qlik Sense alternate states in the app.
- Use alternate states in different objects and sheets.
- Create new expressions using alternate states.

Qlik Sense app: ABC Sales Dashboard v1

Dependencies: Before working in this activity, make sure you complete Activity 2.

Challenge

In the **ABC Sales Dashboard v1** app, create a chart comparison in the *Regional orders* sheet:

- Create a monthly order volume trend line chart to compare two sets based on user selections
- Use set analysis in the charts (in the measure and in the reference line expression):
  - Use a set identifier for the current selections.
  - Use a modifier that selects the value in the *Division* field.
  - Use alternate states combined with set analysis for the comparison trend.
- Show value labels and data points in the line charts

Bonus point:

Create a new sheet called *Set comparison*. Business users will perform sales and margin analysis comparing these metrics between different set of products they select:

- Use two alternate states and two filter panes in the sheet to allow users to select products.
- Display two KPIs with the *Sales* and *Margin%* for each alternate state.
- Visualize in bar charts the sales measure (using the *YearMonth Drill-down* and the *Division* dimensions) comparing the two alternate sets.

Note: Measures and data shown depend on today's date. If you reload your app data, your results might differ from the solution image results.

Hints

- In the chart, use set expressions in each measure to assign different selection states to each calculation.
- The set analysis statement is declared with curly brackets, inside the function.
  - The **$** set identifier represents the current selections.
  - Alternate state names can be used as set identifiers in your set analysis expression.
  - The set modifier specifies the field values. For instance: *Division={'EMEA'}*.
- Outer set expression can be used instead of inner. In that case, the set analysis statement must be place ahead of the function.
- Alternate states can be defined in the app when editing a sheet, using the **Assets panel > Master items** section.

- Assign an alternate state to a sheet object using the object properties panel (under the **Alternate states** section).

Step-by-step guide

Create a comparison chart

1. Drag a filter pane into the *Regional orders*sheet, right below the *"Order volume trend - EMEA"* line chart. Then add *Year* and *Division* as fields.
2. From the **Assets panel > Master items > Alternate states** section, use the **Create new** button to create two new alternate state in the app called *Set1* and *Set2*.
3. In the properties panel of this filter pane, under **Appearance > General,** set the **Title** to "Group1" and under the **Alternate states** section, set the **State** to *Set1*.
4. Copy and paste the filter pane, and in the copy, change the **Title** to "Group2" and set the **State** to *Set2*.
5. Copy and paste the EMEA line chart. You will now have 3 line charts. Then change the title to *Order volume trend - Comparison*.
6. Remove the existing measure and create two measures as follows:

   First measure: set the **Label** to *Group1* and the expression as follows:

   ```
   Count({Set1} DISTINCT OrderID)
   ```

   Second measure: set the **Label** to *Group2* and the expression as follows:

   ```
   Count({Set2} DISTINCT OrderID)
   ```

7. In analysis mode, select the following in the group filters:
   - Group1 filter pane: select *2023* and the *EMEA* division
   - Group2 filter pane: select *2023* and the *South America* and *North America* divisions

   Question: Would it be possible to have dynamic labels in the measures to see what each line represents in the Order volume trend line chart legend?

   Yes, you can use dynamic labels in the filter panes or in the measures of the line chart. Discuss the differences:

   In the filter pane subtitles:

   ```
   =GetFieldSelections( [OrderDate.autoCalendar.Year])&'-'&GetFieldSelec-
   tions( [Division])
   ```

   In the measure labels of each chart:

   ```
   =Concat(distinct {Set1} [OrderDate.autoCalendar.Year],',')&'-'& Concat
   (distinct {Set1} Division,',')
   ```

   ```
   =Concat(distinct {Setp2} [OrderDate.autoCalendar.Year],',')&'-'& Con-
   cat(distinct {Set2} Division,',')
   ```

Bonus point:

8. Create a new sheet in the **ABC Sales Dashboard v1** app, and name it: **Set comparison**.
9. In edit mode, access the sheet properties panel and, under **Alternate states** section, make sure the **State** is set to **<default state>**.
10. From the **Assets panel > Charts** section, drag a **Text & Image** object into the sheet and add the following text under the sheet title: *Select in both category and product filters to analyze two sets side by side.*
11. From the **Assets panel > Charts** section, drag a **Filter pane** in the sheet. Use the *Product* and *Product Category* fields in the filter pane. Then, in the filter pane's properties panel, under **Appearance > Alternate states**, set the **State** to *Set1*.
12. Make sure you display the title in the filter pane and write *Set 1* as the title.
13. Copy and paste the filter pane in the same sheet and set this new filter pane's **State** to *Set2* and change the title to *Set 2*.
14. From the **Assets panel > Master items > Measures** section, drag the *Sales* measure into a free space in the sheet. This measure is represented as a KPI. Then drag also the *Sales CYTD - LYTD* on top to include a secondary measure in the KPI.
15. In the KPI properties panel, set the alternate state to *Set1*. Then copy the KPI and paste it again in the sheet and change the state to *Set2* for the new KPI.
16. From the **Assets panel > Charts** section, drag a new **KPI** in the sheet using the *Margin%* measure and set the alternate state to Set1 in the properties panel. Then copy the Margin% KPI and paste it again in the sheet but change the state to *Set2*.
17. Relocate the KPI and filter pane objects in the sheets, so all the objects with the *Set1* state are displayed on the left and the objects with the *Set2* state are displayed on the right.
18. Create a new bar chart in the sheet using *Year Drill-down* as dimension and the following two expressions:

    ```
    Sum({Set1}Sales)
    ```

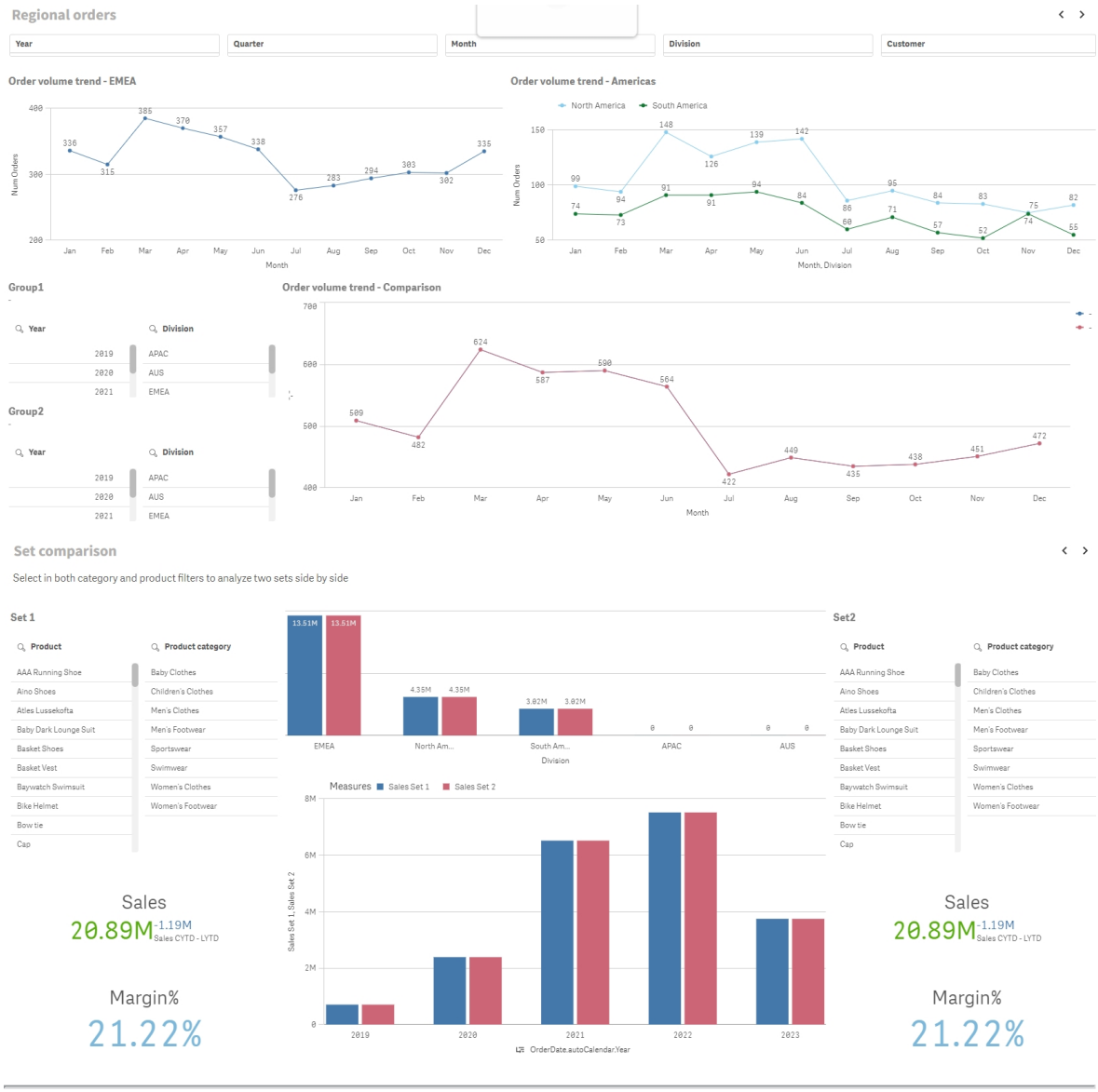
    ```
    Sum({Set2}Sales)
    ```

    By adding the set expressions *{Set1}* and *{Set2}*, the calculation is performed only over the data with the corresponding alternate state selections applied.
19. Rename the new expressions to *Sales Set1* and *Sales Set2* and show the legend at the top.
20. Copy and paste the chart to duplicate it. In the new bar chart, remove the existing dimension, then include *Division* as the new dimension and hide legend and Y-axis.

Analyzing the sheet

21. In analysis mode, make selections in *Sportwear* category using the filter pane in Alternate state Set1, then in the other filter pane, select all Men's product categories and compare both charts. Compare with the solution image.

Solution


Regional orders
Year
Quarter
Month
Division
Customer
Order volume trend - EMEA
Num Orders
Month
Order volume trend - Americas
North America
South America
Num Orders
Month, Division
Group1
Year
Division
2019
2020
2021
APAC
AUS
EMEA
Group2
Year
Division
2019
2020
2021
APAC
AUS
EMEA
Order volume trend - Comparison
Month
Set comparison
Select in both category and product filters to analyze two sets side by side
Set 1
Product
Product category
AAA Running Shoe
Aino Shoes
Atles Lussekofta
Baby Dark Lounge Suit
Basket Shoes
Basket Vest
Baywatch Swimsuit
Bike Helmet
Bow tie
Cap
Baby Clothes
Children's Clothes
Men's Clothes
Men's Footwear
Sportswear
Swimwear
Women's Clothes
Women's Footwear
EMEA
North Am...
South Am...
APAC
AUS
Division
Measures
Sales Set 1
Sales Set 2
Sales Set 1, Sales Set 2
OrderDate.autoCalendar.Year
Set2
Sales
20.89M
-1.19M
Sales CYTD - LYTD
Margin%
21.22%
Sales
20.89M
-1.19M
Sales CYTD - LYTD
Margin%
21.22%

(Optional) Challenge 4: Validating set expressions

Summary:

- Understand set analysis syntax and be able to fix incorrect syntax.

Qlik Sense app: ABC Sales Dashboard v1

Challenge

Review the expressions using set analysis, identify possible errors, and suggest corrections.

Note that these expressions are using the data fields and values from the **ABC Sales Dashboard v1** app.

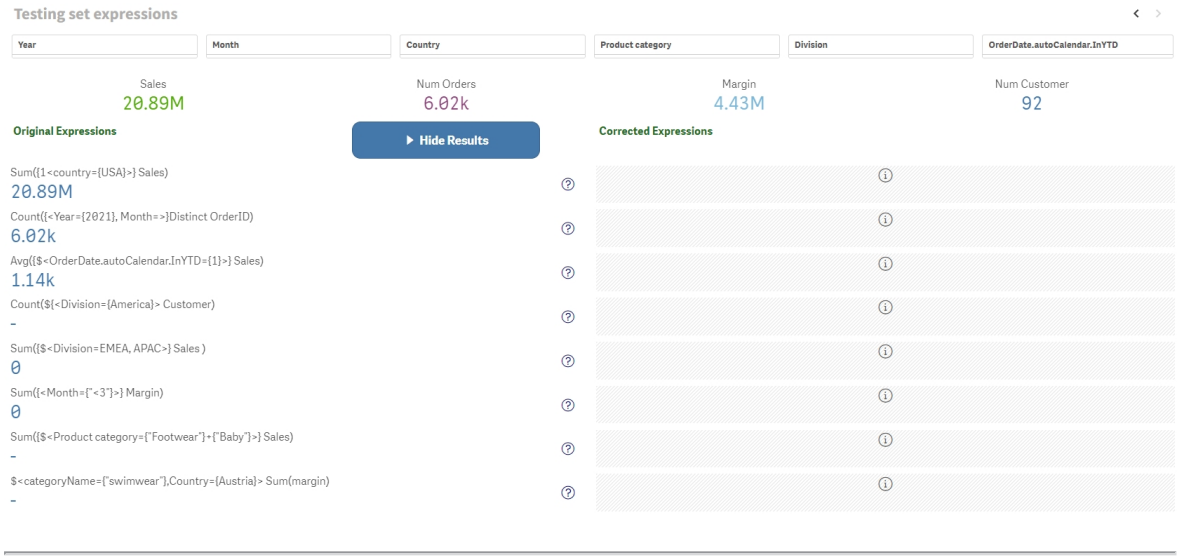
Tip: There is a hidden sheet in the app with these formulas. Set the *vSetAnalysis* variable to 1 to make the sheet visible. Then use the button to show or hide the solution.

Hints

- Review the table below and check whether the set analysis syntax in these expressions could be used in the app.
- Consider the answers to these questions as they relate to each expression:
- Is the expression correct if it is used in the app? If not, could you edit it so it is a correct expression for the app?
- Please be aware that not all the expressions have syntax errors.

ID	Expression
1	`Sum({1<country={USA}>} Sales)`
2	`Count({<Year={2021}, Month=>}Distinct OrderID)`
3	`Avg({$<OrderDate.autoCalendar.InYTD={1}>} Sales)`
4	`Count(${<Division={America}> distinct Customer)`
5	`Sum({$<Division=EMEA, APAC>} Sales)`
6	`Sum({<Month={"<3"}>} Margin)`
7	`Sum({$<Product category={“Footwear”}+{“Baby”}>} Sales)`
8	`$<categoryName={"swimwear"},Country={Austria}> Sum(margin)`

Solution


Testing set expressions
Year
Month
Country
Product category
Division
OrderDate.autoCalendar.InYTD
Sales
20.89M
Num Orders
6.02k
Margin
4.43M
Num Customer
92
Original Expressions
Hide Results
Corrected Expressions
Sum({1<country={USA}>} Sales)
20.89M
Count({<Year={2021}, Month=>}Distinct OrderID)
6.02k
Avg({$<OrderDate.autoCalendar.InYTD={1}>} Sales)
1.14k
Count($({<Division={America}> Customer)
-
Sum({$<Division=EMEA, APAC>} Sales)
0
Sum({<Month={"<3"}>} Margin)
0
Sum({$<Product category={"Footwear"}+{"Baby"}>} Sales)
-
$<categoryName={"swimwear"},Country={Austria}> Sum(margin)
-

(Optional) Activity 5: Combining set analysis and bookmarks

Summary:

- Explore and work with bookmarks and set analysis.
- Use bookmarks to simplify set analysis syntax.

Qlik Sense app: ABC Sales Dashboard v1

Challenge

In the **Dashboard** sheet of the **ABC Sales Dashboard v1** app, create a new visualization displaying the monthly sales trend comparing EMEA vs Americas (includes *South America* and *North America*).

Create two bookmarks and then, create two measures in the chart using the them in set analysis (one for the EMEA division and one for the Americas divisions).

Bonus point:

- Make sure the line chart you create is also affected by the user selections. Use a set operator in set analysis to achieve this.

Hints

- Line charts are good for representing trends over time. Use a line chart with *Month* as a dimension.
- Bookmark names can be used as set identifiers in set analysis, for instance, *Sum({Americas} Sales).*
- Outer set expression can be used instead of inner. In that case, the set analysis statement must be place ahead of the function.

Step-by-step guide

1. In the **ABC Sales Dashboard v1** app, open the **Dashboard** sheet in edit mode.
2. Click **Extend sheet** to leave some free space at the bottom of the sheet and add a **Text & image** object with the following text:

   Sales trend

   How are the sales trends going in the business regions?
3. Click **Done editing** to enable the analysis mode. Now you can perform selections and searches.
4. Search for *EMEA*, and select the value in the *Division* field. Then click the **> Bookmarks** button and select **Create new bookmark**.
5. Add a title, *EMEA*, and click **Create**. Click the **> Bookmarks** button again to close the panel.
6. In the selection bar, select the *Division* field, clear *EMEA* and select *North America* and *South America* values.
7. Click the **> Bookmarks** button and select **Create new bookmark**. Then add a title, *Americas*, and click **Create**.

8. Click the **> Bookmarks** button again to close the panel.
9. Add a **Line chart** below the Text&Image object, in a free space on the sheet. Then select *Month* as a dimension in the new line chart.
10. Using the properties panel of the new line chart, add two measures, with the following expressions:
    - *Americas*:

      ```
      Sum({Americas} Sales)
      ```

    - *EMEA*:

      ```
      Sum({EMEA} Sales)
      ```

11. Under the **Sorting** section, make sure *Month* appears as the first item and it is sorted numerically and ascending.
12. Under the **Appearance** > **Presentation** section, use the **Area** style, check **Show data points**, and set **Value labels** to **Auto**.
13. Add a title to the chart, *Sales trend Americas vs EMEA*.
14. In analysis mode, use the smart search to select the *Q2* quarter. Notice how the line chart does not change with the selection. Discuss how the line chart can change based on the user selections.

Bonus point:

Use a set operator in the set analysis expressions:

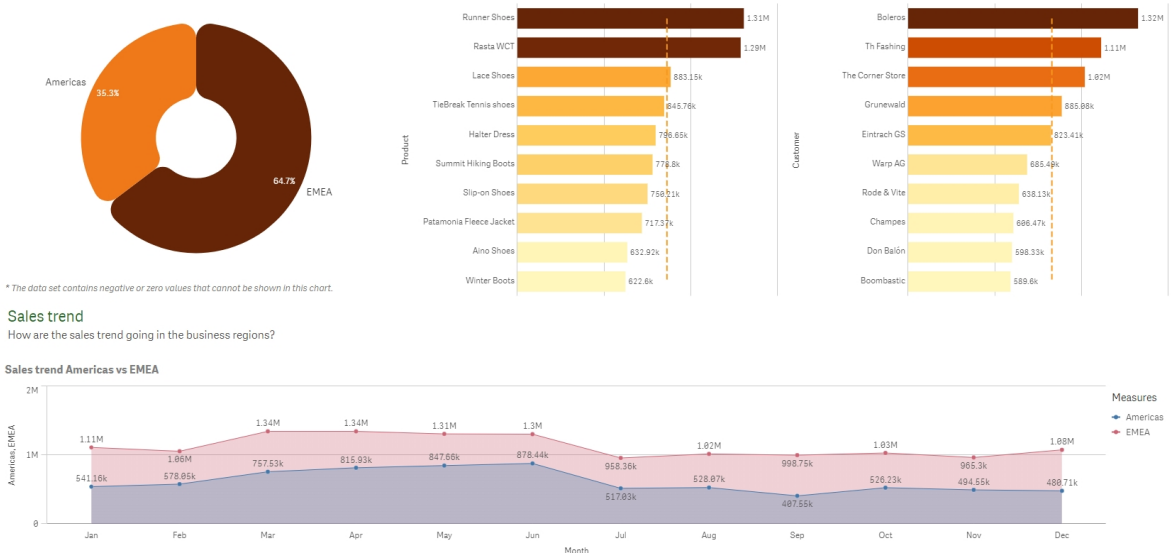
15. In edit mode, modify the *Americas* measure in the line chart so the expression is as follows:

    ```
    Sum({Americas * $} Sales)
    ```

16. Repeat the previous step for the *EMEA* measure, so the expression for this measure is as follows:

    ```
    Sum({EMEA * $} Sales)
    ```

Solution



Sales trend

How are the sales trend going in the business regions?